What Is a Taxon, What Is a Clade, and How Does Tree Surgery Work on a 14B LLM?

14.7B Weights = 48 Layers × 13,824 MLP Units/Layer | 1 Taxon (Leaf) = 1 MLP Unit (Row i of W_gate, Row i of W_up, Col i of W_down = 15,360 floats)

STEP 1 • ONE TAXON

14B Weights vs. 13,824 Units: What Is 1 Taxon?

1 Taxon = MLP Hidden Unit #i (Row i of W_gate + Row i of W_up + Col i of W_down)

'14B' = 14.7B Scalar Weights (NOT 14B Neurons!)

48 Transformer Layers

Total = 14.7B scalar floats

↓

1 MLP Block (Layer l)

212.3M floats per layer
(× 48 layers = 10.2B MLP floats)

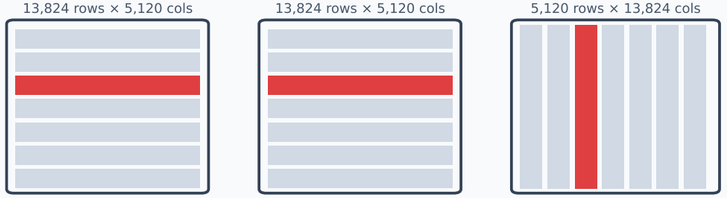
↓

13,824 MLP Units / Layer

Each Unit i = 15,360 floats
(13,824 × 15,360 = 212.3M)

Anatomy of 1 Taxon (Unit #i) Inside Layer l's SwiGLU MLP

Unit i computes: a_i = SiLU(Row_i(W_gate)·x) × (Row_i(W_up)·x) → adds a_i · Col_i(W_down)



Red = Row i (1×5,120) (Gate: When to fire)

Red = Row i (1×5,120) (Up: Feature value)

Red = Col i (5,120×1) (Down: Output vector)

TAXON #i (1 Leaf) = [Row_i(W_gate) | Row_i(W_up) | Col_i(W_down)^T] (15,360 floats)

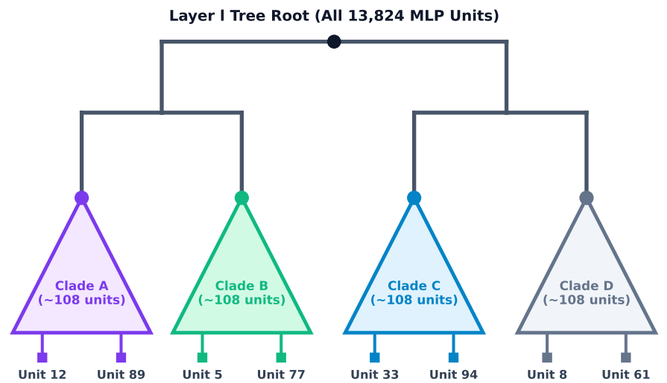
Summary of Step 1: We build ONE tree per Transformer layer (48 trees total). In Layer l's tree, there are N = 13,824 taxa (leaves). Each taxon i (i = 1..13,824) is ONE MLP Hidden Unit — consisting of Row i of W_gate, Row i of W_up, and Column i of W_down (15,360 weights).

STEP 2 • ONE CLADE

What Is the Tree & What Is a Clade in the Matrices?

13,824 leaves per layer tree; 1 Clade (~108 leaves) = 108 rows of W_gate/up & 108 cols of W_down

In Layer l: 13,824 Taxa (MLP Units) → 13,824×13,824 Distance Matrix → Binary Tree with 13,824 Leaves

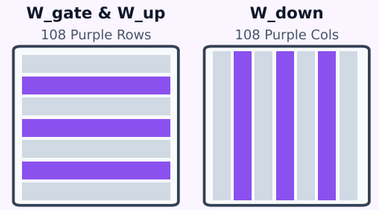


1 Leaf = 1 MLP Unit (15,360 floats) | 1 Clade = Subtree of ~108 Related Units

Clade A Index Set →

What 1 Clade (e.g. Clade A) Is in the Matrices

Clade A = 108 Unit IDs, e.g. C_A = {12, 89, 401, ...}



• A Clade is a matched set of ~108 Rows in W_gate & W_up plus the matching ~108 Columns in W_down (~1.66M floats).
• Together, those 108 units form a self-contained sub-MLP circuit that detects a pattern and writes to the residual stream!

Summary of Step 2: We compute the 13,824 × 13,824 cosine distance matrix between the 13,824 units in Layer l and infer a binary tree. Cutting the tree at ~128 subtrees partitions the 13,824 units into ~128 Clades (~108 units/clade × 48 layers = 6,171 clades across the 14B model).

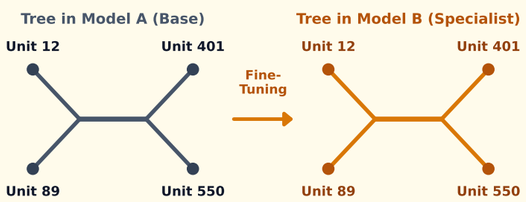
STEP 3 • TWO MODELS

Comparing Base Tree vs. Specialist Tree (Same Leaf IDs)

Why Quartet Violations catch units swapping partners while ignoring rigid block rotation

Case A: Clade Drifts / Rotates as a Rigid Block

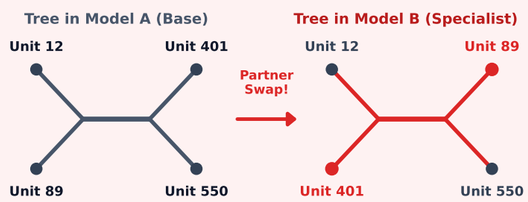
Base Tree vs. Specialist Tree Have the EXACT Same Quartet Partners



• All 4 units changed their 15,360 floats (||W_B − W_A|| is large), which fools standard weight-norm & TIES methods.
• But Unit 12 is still paired with Unit 89 in both trees!
• Quartet Violation = 0 → No internal circuit rewiring.

Case B: Units Swap Partners Between Models!

Quartet Violation Detects Clades That Rewired Their Function



• In Model A (Base), Unit 12 was partnered with Unit 89.
• In Model B (Specialist), Unit 12 broke away to pair with Unit 401!
({12, 89}|{401, 550} → {12, 401}|{89, 550})
• High Quartet Violation Score → This clade rewired its circuit!

Summary of Step 3: Because Unit #12 in Model A and Unit #12 in Model B have the exact same leaf ID, we compare Layer l's tree in Model A against Layer l's tree in Model B. Clades whose units swapped tree partners (Quartet Violations) are the sub-circuits that rewired!

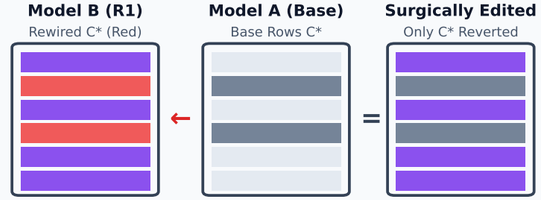
STEP 4 • SURGERY

Exact Row & Column Slicing on W_gate, W_up, W_down

Swapping the 108 rows & columns of a rewired Clade C* between models (Zero Training)

Surgery 1: Clade Knock-Out (Causal Ablation)

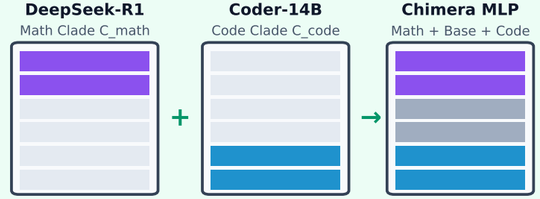
Overwrite ONLY Clade C*'s 108 Rows/Cols with Base Weights



Exact Matrix Operation for a Rewired Clade C* = {12, 89, ...}:
W_gate[C*, :] ← W_gate_base[C*, :] (108 rows reverted)
W_up[C*, :] ← W_up_base[C*, :] (108 rows reverted)
W_down[:, C*] ← W_down_base[:, C*] (108 cols reverted)
All other 13,716 units in Layer l stay 100% untouched!

Surgery 2: Clade Grafting (Zero-Training Chimera)

Transplant Disjoint Specialist Clades into One MLP Matrix



Exact Matrix Operation (with C_math ∩ C_code = ∅):
• Start with Base W_gate, W_up, W_down
• Copy rows/cols C_math from DeepSeek-R1 (purple)
• Copy rows/cols C_code from Qwen-Coder (blue)
→ Both specialist sub-circuits coexist without colliding!

Summary of Step 4: 'Tree Surgery' means using a clade's leaf IDs C = {i_1, ..., i_108} as an index mask to swap the matching 108 rows of W_gate, 108 rows of W_up, and 108 columns of W_down between models — transplanting an entire co-adapted multi-neuron circuit as an intact unit!